

Formulas for Predicted y value

Train/Test Split

I split the dataset into training and testing sets 80/20. Fit models using training set, while testing set used to determine performance of the model. Fixed random state = 42 for reproducibility.

OLS

$$y_{\text{hat}} = \beta_0 + \sum (\beta_j * x_j)$$

Ridge Regression

$$y_{\text{hat}} = \beta_0 + \sum (\beta_j x_j)$$

Where β is estimated using a penalty term $\lambda \sum \beta_j^2$

Used 4-fold cross validation to find optimal lambda from a set of lambdas [0.001, 0.01, 0.1, 1, 2, 4, 7, 10, 50, 100].

I chose the lambda = 4 with the lowest validation error.

Nadaraya-Watson Kernel Regression

$$y_{\text{hat}}(x) = \frac{\sum K((x - x_i)/h) y_i}{\sum K((x - x_i)/h)}$$

Used 4-fold cross validation to find optimal bandwidth h from a set of [100, 200, 300, 400, 500, 750, 1000, 1500, 2000, 3000].

I chose h = 400 with the lowest validation error.

Test MSE for each model (copied from my code output)

OLS TEST MSE: 1754318687330.6638

Ridge TEST MSE: 1763262838872.424

Kernel Regression TEST MSE: 3807823878474.287

OLS had the lowest test MSE with the best prediction performance. Ridge regression performed slightly worse, so regularization did not help much in improvement. Kernel regression had the highest error by a large margin because only using one feature to predict housing prices probably led to an underfitting of the data.

Code link:

https://colab.research.google.com/drive/1WRtShGVD-mgFnz7uESmdAJ74jeNWhv_i?usp=sharing